

1 Rollout

policy

π_t



transitions

GAE

A_r, A_{c_k}

episode cost

$\bar{J}_k$

2 Flow inputs

state clusters



transition edges



$\hat{F}$
flow



advantage



C_k
costs

3 Set constraints

Soft constraints

all cost budgets



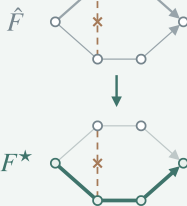
Hard constraints



remove edges

4 Entropic projection

One flow solve



target flow

is not executed

prices go to the actor

5 Priced policy update

Safety prices

clip + EMA

Observed-cost feedback

cost - budget

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO/TRPO $\rightarrow \pi_{t+1}$

observed episode costs

GAE advantages

next rollout